CoT-UQ

CoT-UQ[122]

☐ Step 1: Reasoning Extraction

The model generates an explicit step-by-step reasoning process using Chain-of-Thought prompting, together with a final answer.

☐ Step 2: Keywords Extraction

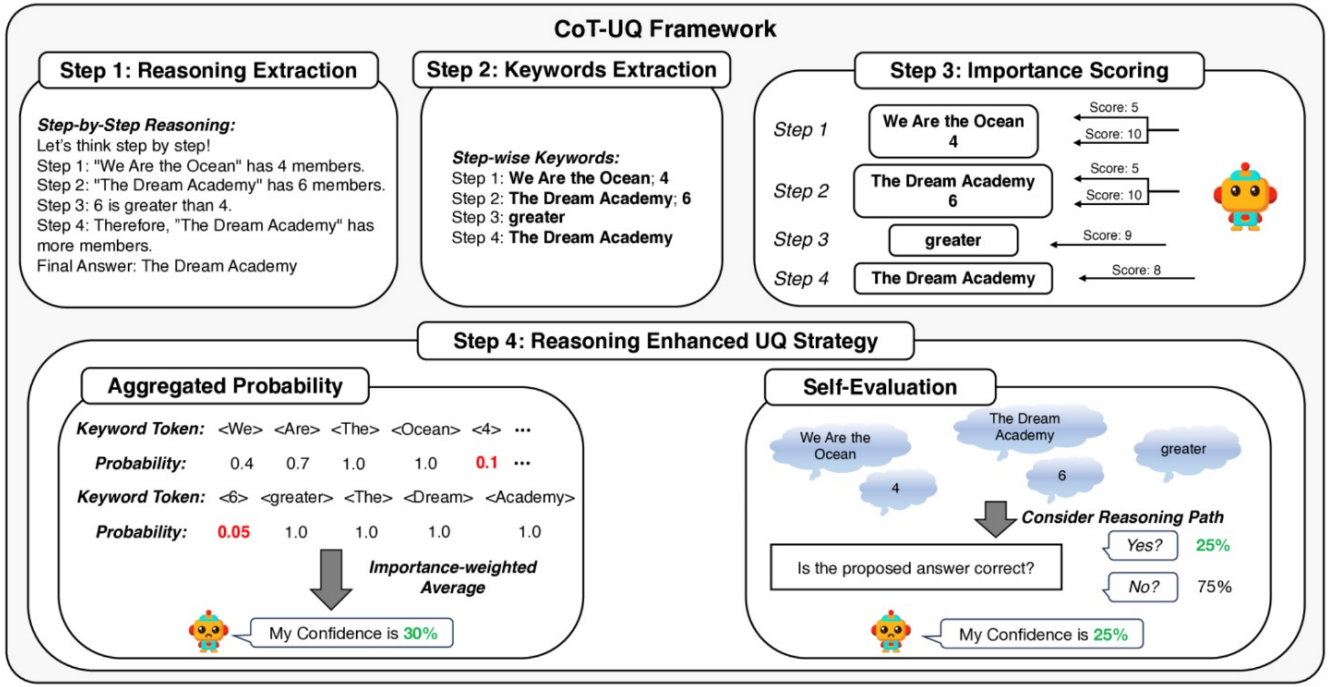
Key tokens (e.g., entities, numbers, logical relations) are extracted from each reasoning step to retain only the most informative reasoning components.

☐ Step 3: Importance Scoring

Each extracted keyword is assigned an importance score indicating its contribution to deriving the final answer.

☐ Step 4: Reasoning Enhanced UQ Strategy

The extracted keywords and their importance scores are integrated into existing uncertainty estimation strategies(AP & SE) to produce calibrated response-wise uncertainty.



[122] CoT-UQ: Improving Response-wise Uncertainty Quantification in LLMs with Chain-of-Thought, ACL 2025

CoT-UQ[122]

- **Task Type:** Opening-end/Math/Code generation

- **Introduction:** CoT-UQ integrates Chain-of-Thought reasoning into uncertainty quantification by extracting step-wise keywords, assigning importance scores, and aggregating their token-level probabilities to form a response-wise confidence score. It quantifies reasoning uncertainty and provides a structured way to incorporate internal reasoning signals into existing UQ strategies.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Reasoning/Token-level	White-box	Post-hoc	Medium	Ranking

[122] CoT-UQ: Improving Response-wise Uncertainty Quantification in LLMs with Chain-of-Thought, ACL 2025